LF Denoising Network

Alias	Layer	Operations	Outputs' Shape
Primary Feature Extraction	LF To ViewStack	concat	(225,24,24,1)
	PFE_1	conv2d (n32 k3 s1)	(225,24,24,32)
	PFE_2	conv2d (n32 k3 s1 dila1)-> conv2d (n32 k3 s1 dila2)-> conv2d (n32 k3 s1 dila4)->concat	(225,24,24,96)
	PFE_3	conv2d (n32 k3 s1)	
	PFE_4	conv2d (n32 k3 s1)->relu-> conv2d (n32 k3 s1)	(225,24,24,32)
		add (PFE_3,PFE_4)	
Dual-attention Module	VA1 (View Attention) Input	transpose	
	VA1_FC1	GlobalMeanPooling ->conv2d (n16 k3 s1) -> conv2d (n16 k1 s1)	
	VA1_FC2	GlobalMaxPooling ->conv2d (n16 k3 s1) -> conv2d (n16 k1 s1)	
	VA1_value	merge(FC1,FC2)->sigmoid	
		multiply (VA1_value,VA1)	
		conv2d (n32 k3 s1)->relu-> conv2d (n32 k3 s1)	
	VA1 Output	conv2d (n32 k3 s1)->relu-> conv2d (n32 k3 s1)	
	VA2 Input	merge (VA1 Input, VA1 Output)	(32,24,24,225)
	VA2_FC1	GlobalMeanPooling ->conv2d (n16 k3 s1) -> conv2d (n16 k1 s1)	
	VA2_FC2	GlobalMaxPooling ->conv2d (n16 k3 s1) -> conv2d (n16 k1 s1)	
	VA2_value	merge(FC1,FC2)->sigmoid	
		multiply (VA1_value,VA1)	
		conv2d (n32 k3 s1)->relu-> conv2d (n32 k3 s1)	
	VA2 Output	conv2d (n32 k3 s1)->relu-> conv2d (n32 k3 s1)	
	VA results	merge (VA1 Input, VA1 Output,VA2 Output)	
	CA1 (Channel Attention) Inpu	transpose	
	CA1_FC1	GlobalMeanPooling ->conv2d (n16 k3 s1) -> conv2d (n16 k1 s1)	
	CA1_FC2	GlobalMaxPooling ->conv2d (n16 k3 s1) -> conv2d (n16 k1 s1)	
	CA1_value	merge(FC1,FC2)->sigmoid	
		multiply (CA1_value,CA1)	
		conv2d (n32 k3 s1)->relu-> conv2d (n32 k3 s1)	
	CA1 Output	conv2d (n32 k3 s1)->relu-> conv2d (n32 k3 s1)	
	CA2 Input	merge (CA1 Input, CA1 Output)	(225,24,24,32)
	CA2_FC1	GlobalMeanPooling ->conv2d (n16 k3 s1) -> conv2d (n16 k1 s1)	
	CA2_FC2	GlobalMaxPooling ->conv2d (n16 k3 s1) -> conv2d (n16 k1 s1)	
	CA2_value	merge(FC1,FC2)->sigmoid	
		multiply (CA1_value,CA1)	
		conv2d (n32 k3 s1)->relu-> conv2d (n32 k3 s1)	
	CA2 Output	conv2d (n32 k3 s1)->relu-> conv2d (n32 k3 s1)	
	CA results	merge (CA1 Input, CA1 Output,CA2 Output)	
Feature Fusion & Output	F_0	concat(VA results,CA results)-> conv2d (n16 k1 s1)	
	F_1	conv2d (n8 k3 s1)->relu-> conv2d (n16 k3 s1)	
	F_2	merge(F_0,F_1)	(225,24,24,16)
	F_3	conv2d (n8 k3 s1)->relu-> conv2d (n16 k3 s1)	
		merge(F_2,F_3)	
	Output	conv2d (n1 k3 s1)	(225,24,24,1)
	Conver to SAIs	concat	(1,360,360,1)

LF De-aliasing Network

Alias	Layer	Operations	Outputs' Shape
Input	conver to Macron	concat	(1,360,360,1)
	PFE_A1	conv2d (n64 k15 s15)	(1, 24, 24, 64)
	F1_S1	conv2d (n14400 k1 s1)->subpixel(r=15)	
	F1_S2	concat (PFE_S1,F1_S1)->conv2d (n64 k3 s1 dila15)->Relu	
	F1_S3	merge(PFE_A1,F1_A2)	(1, 360, 360, 64)
	F1_S1_1	conv2d (n14400 k1 s1)->subpixel(r=15)	
Angular Branch	F1_S2_1	concat (PFE_S1,F1_S1_1)->conv2d (n64 k3 s1 dila15)->Relu	
	F1_S3_1	merge(PFE_A1,F1_A2_1)	
	F2_A1	conv2d (n64 k15 s15)	
	F2_A2	concat (F1_A3,F2_A1)->conv2d (n64 k3 s1)->Relu	(1, 24, 24, 64)
	F2_A3	merge(F1_A3,F2_A2)	
	PFE_S1	conv2d (n64 k3 s1 dila15)	(1,360,360,64)
	F1_A1	conv2d (n64 k15 s15)	
	F1_A2	concat (PFE_A1,F1_A1)->conv2d (n64 k3 s1)->Relu	
	F1_A3	merge(PFE_A1,F1_A2)	(1, 24, 24, 64)
	F1_A1_1	conv2d (n64 k15 s15)	
Spatial Branch	F1_A2_1	concat (PFE_A1,F1_A1_1)->conv2d (n64 k3 s1)->Relu	
	F1_A3_1	merge(PFE_A1,F1_A2_1)	
	F2_S1	conv2d (n14400 k1 s1)->subpixel(r=15)	
	F2_S2	concat (F1_S3,F2_S1)->conv2d (n64 k3 s1 dila15)->Relu	(1, 360, 360, 64)
	F2_S3	merge(F1_S3,F2_S2)	
Feature Fusion	L1_A	concat(F1_A3_1,F2_A3)->conv2d(n64 k1 s1)->Relu->conv2d(n14400 k1 s1)->subpixel	
	L1_S	concat(F1_S3_1,F2_S3,L1_A)->conv2d (n64 k3 s1 dila15)->Relu	(1, 360, 360, 64)
	L2_S	merge(PFE_S1,L1_S)	
Output	Output	conv2d (n25 k3 s1 dila15)->subpixel (r=5)->conv2d(n1 k1 s1)	(1, 1800, 1800, 1)

3D Reconstruction Network

Alias	Layer	Operations	Outputs' Shape
Input	Inputlayer		
Spatai Feature Extraction	reshape	Space to Depth	(1, 120, 120, 225)
	l0	conv2d(n128 k7 s1)	(1, 120, 120, 128)
	l1	subpixel(r=2)->conv2d(n64 k3 s1)	(1, 240, 240, 64)
	l2	subpixel(r=2)->conv2d(n32 k3 s1)	(1, 480, 480, 64)
Feature Fusion	l0	conv2d(n32 k3 s1)->InstanceNorm->LeakyRelu	(1, 480, 480, 32)
U-Net (Encoder)	En_0	conv2d(n64 k3 s1)->InstanceNorm->LeakyRelu	(1, 480, 480, 64)
	l0	conv2d(n128 k1 s1)->InstanceNorm->LeakyRelu	
	l1	conv2d(n32 k3 s1)->InstanceNorm->LeakyRelu	
	l2	conv2d(n32 k3 s1)->InstanceNorm->LeakyRelu	
	l3	conv2d(n64 k3 s1)->InstanceNorm->LeakyRelu	
	l4	concat(l1,l2,l3)->InstanceNorm	
		merge(l0,l4)->LeakyRelu->InstanceNorm	(1, 480, 480, 128)
	En_1	Maxpooling (k3 s2)	(1, 240, 240, 128)
	l0	conv2d(n256 k1 s1)->InstanceNorm->LeakyRelu	
	l1	conv2d(n64 k3 s1)->InstanceNorm->LeakyRelu	
	l2	conv2d(n64 k3 s1)->InstanceNorm->LeakyRelu	
	l3	conv2d(n128 k3 s1)->InstanceNorm->LeakyRelu	
	l4	concat(l1,l2,l3)->InstanceNorm	(1, 240, 240, 256)
	En_2	Maxpooling (k3 s2)	(1, 120, 120, 256)
	l0	conv2d(n512 k1 s1)->InstanceNorm->LeakyRelu	
	l1	conv2d(n128 k3 s1)->InstanceNorm->LeakyRelu	
	l2	conv2d(n128 k3 s1)->InstanceNorm->LeakyRelu	
	l3	conv2d(n256 k3 s1)->InstanceNorm->LeakyRelu	
	l4	concat(l1,l2,l3)->InstanceNorm	(1, 120, 120, 512)
	En_3	Maxpooling (k3 s2)	(1, 60, 60, 512)
	l0	conv2d(n512 k1 s1)->InstanceNorm->LeakyRelu	
	l1	conv2d(n128 k3 s1)->InstanceNorm->LeakyRelu	
	l2	conv2d(n128 k3 s1)->InstanceNorm->LeakyRelu	
	l3	conv2d(n256 k3 s1)->InstanceNorm->LeakyRelu	
	l4	concat(l1,l2,l3)->InstanceNorm	(1, 60, 60, 512)
	En_4	Maxpooling (k3 s2)	(1, 30, 30, 512)
	l0	conv2d(n512 k1 s1)->InstanceNorm->LeakyRelu	
	l1	conv2d(n128 k3 s1)->InstanceNorm->LeakyRelu	
	l2	conv2d(n128 k3 s1)->InstanceNorm->LeakyRelu	
	l3	conv2d(n256 k3 s1)->InstanceNorm->LeakyRelu	
	l4	concat(l1,l2,l3)->InstanceNorm->Maxpooling (k3 s2)	(1, 15, 15, 512)
U-Net (Middle)	Middle	Upsamling	(1, 30, 30, 512)
		concat(En_4,Middle)	(1, 30, 30, 1024)
U-Net (Decoder)	DE_1	con2d(n512 k3 s1)-> LeakyRelu->InstanceNorm->Upsamling	(1, 60, 60, 512)
	DE_2	concat(En_3,DE_1)->con2d(n512 k3 s1)-> LeakyRelu->InstanceNorm->Upsamling	(1, 120, 120, 512)
	DE_3	concat(En_2,DE_2)->con2d(n256 k3 s1)-> LeakyRelu->InstanceNorm->Upsamling	(1, 240, 240, 256)
	DE_4	concat(En_1,DE_3)->con2d(n128 k3 s1)-> LeakyRelu->InstanceNorm->Upsamling	(1, 480, 480, 128)
	DE_5	concat(En_0,DE_4)->con2d(n161 k3 s1)-> LeakyRelu->InstanceNorm->Upsamling	(1, 720, 720, 161)
Refine & Output	conv_final	conv2d(slices k3 s1)	(1, 720, 720, 161)